

// LE BONHOMME PHARMA • MONTRÉAL • OPEN SCIENCE

FlexAID Δ S

Benchmarking an entropy-driven
molecular docking engine.

$$\Delta G = \Delta H - T\Delta S$$

Apache 2.0 / C++26 • Python • PyMOL / github.com/LeBonhommePharma

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One equation, three benchmarks, an engine you can rerun.

- 01 First principles — why binding is a free energy, not a score
- 02 The benchmark — the same code ranks 4th, 2nd, and 1st
- 03 Real atoms — binding modes and pose populations, live
- 04 The engine — architecture, scoring, and what changed
- 05 The kernel, the roadmap, and the ask

$$\Delta G = \Delta H - T\Delta S$$

- ΔH — how good the best pose is
- $T\Delta S$ — how many good poses there are
- ΔG — what actually binds

// FIRST PRINCIPLES

Binding is a free energy,
not a score.

$$\Delta\tilde{G} = \Delta\tilde{H} - T \cdot \Delta\tilde{S}$$

$\Delta\tilde{H}$ · enthalpy

The weighted mean contact score of the mode. How good the best pose is — the only thing classical docking reports.

$\Delta\tilde{S}$ · entropy

Shannon entropy of the occupied basin, $-\sum p \cdot \ln p$. How many good poses there are — the term everyone drops.

$\Delta\tilde{G}$ · free energy

The ranking objective FlexAIDΔS actually sorts on. What binds at finite temperature, not just the deepest well.

// THE PROBLEM

The lowest-energy pose is not always the answer.

Ranking on min(CF) conflates a real, narrow binding funnel with a broad, flat false minimum — many structurally diverse poses that happen to score near the same value. Worse, the genetic algorithm amplifies it: selection pressure crowds chromosomes into the low-CF region long before the physically correct pose has been resolved.

What min(CF) sees

One number, one pose. It cannot tell a deep funnel from a flat plateau — so it will confidently elect a false minimum that no crystal ever occupies.

argmin over poses → a point estimate

What the ensemble sees

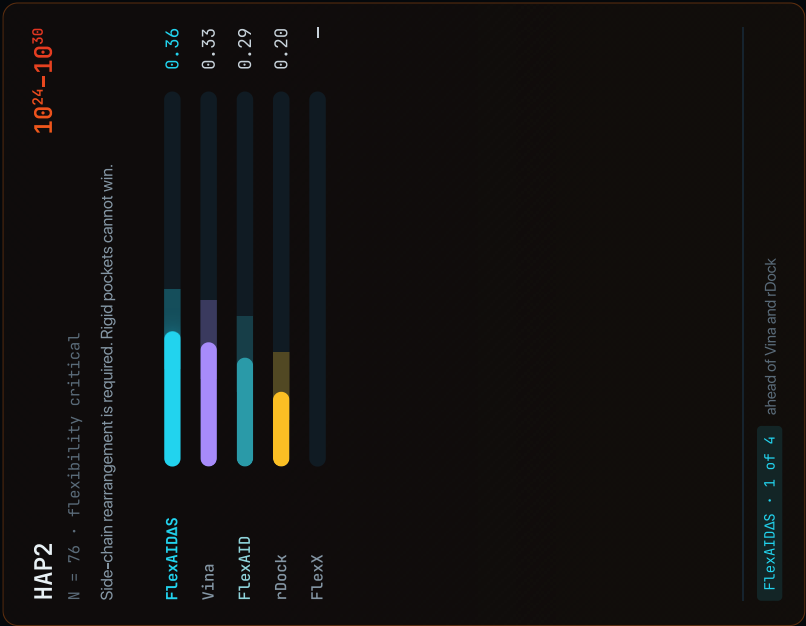
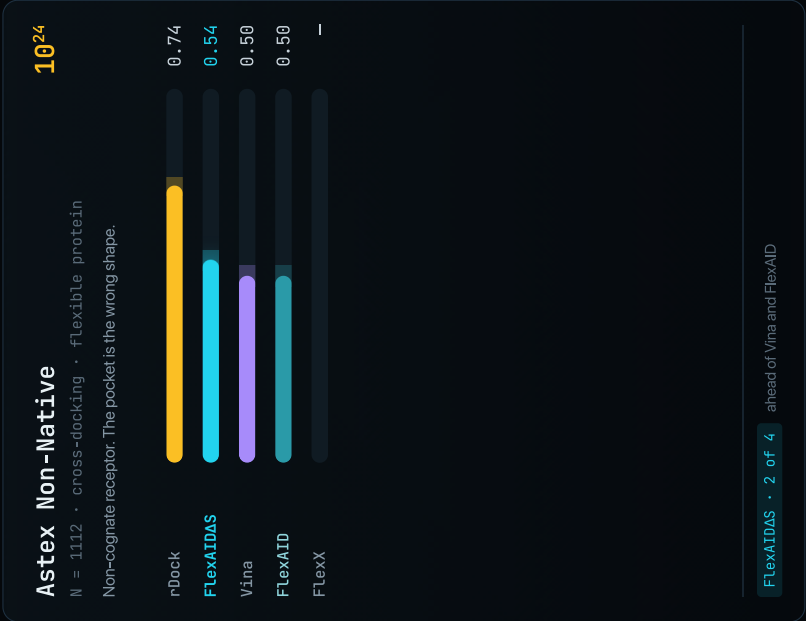
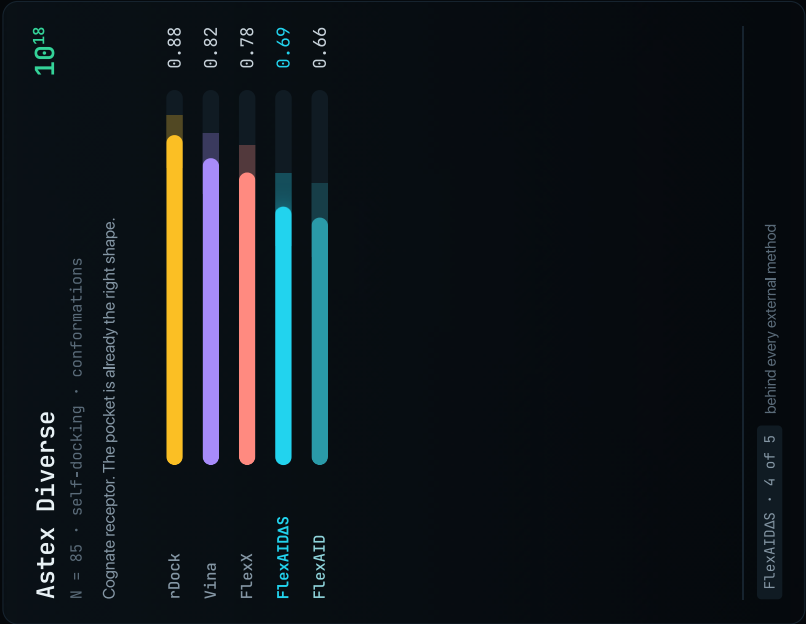
The whole GA-sampled population as a distribution. A mode is favoured when it is both low-energy and densely populated — enthalpy and entropy together.

$$\bar{G} = \bar{H} - T \cdot \bar{S} \text{ over the mode}$$

// BENCHMARK · RANKING IS A DISTRIBUTION PROBLEM

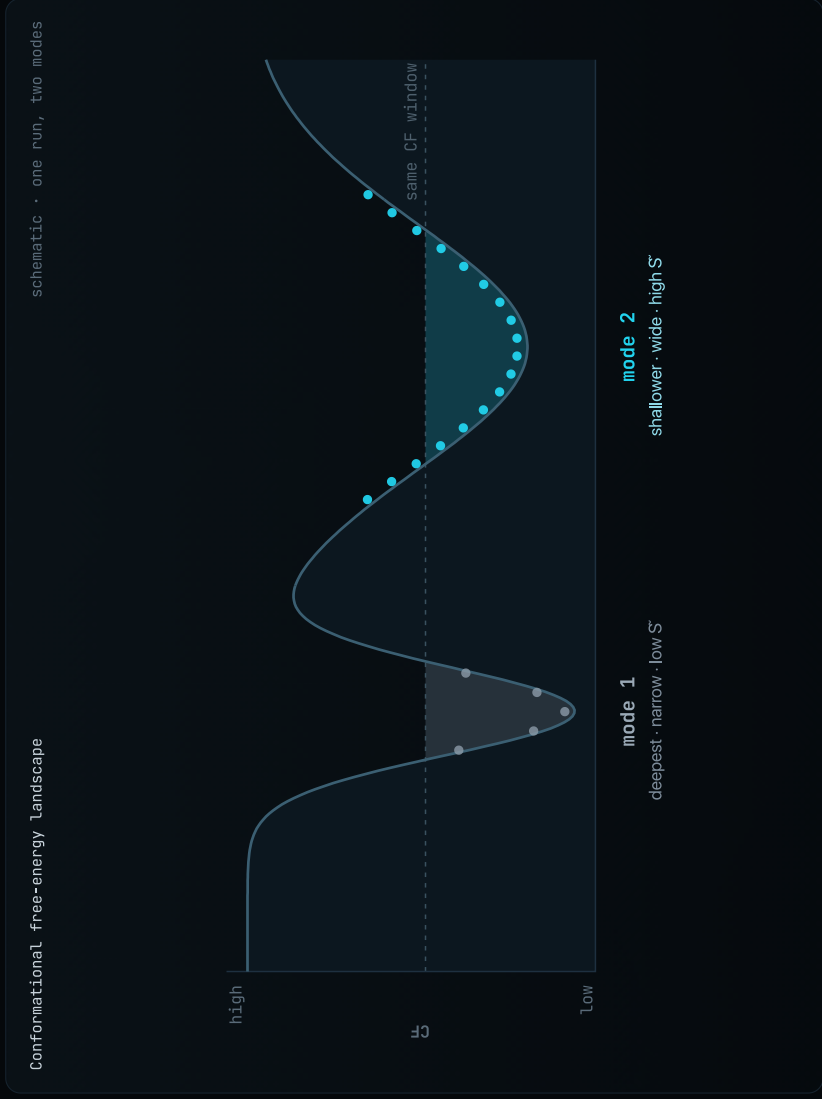
A pose is a sample. Binding is a distribution.

median success · RMSD < 2.0 Å
top-10 poses · 10 runs × 2M evals
band = Wilson 95% CI from N



// WHY ENTROPY

Enthalpy elects the deepest pose.
Free energy elects the **populated** basin.



Conformational free-energy landscape

schematic · one run, two modes

PER-POSE, FROM THE ENSEMBLE

$$Z = \sum_i \exp(-CF_i/T)$$

$$P_i = \exp(-CF_i/T) / Z$$

$$\Delta \tilde{H} = \sum_i P_i \cdot CF_i$$

$$\Delta \tilde{S} = -\sum_i P_i \cdot \ln P_i$$

$$\Delta \tilde{G} = \Delta \tilde{H} - T \cdot \Delta \tilde{S}$$

partition function

Boltzmann weight

weighted enthalpy

Shannon entropy

the objective

$\Delta \tilde{H}$ alone

"Mode 1 is preferred — it contains the pose of lowest energy." Steep well, few poses.

$$\Delta \tilde{G} = \Delta \tilde{H} - T \Delta \tilde{S}$$

"Mode 2 is preferred — it contains multiple poses of favourable energy." — 3Dsig 2017

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// BINDING MODES VS THE CRYSTAL
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The populated mode **is** the crystal pose. 0.52 Å.
The runner-up is 3.71 Å off.

10F1 · 789 pocket atoms · 10 docked poses + crystal

bond orders from RDKit



● **mode 1** matches crystal

0.52 Å from crystal

8 poses · best CF -175.83
RMSD to crystal 0.52-2.73 Å

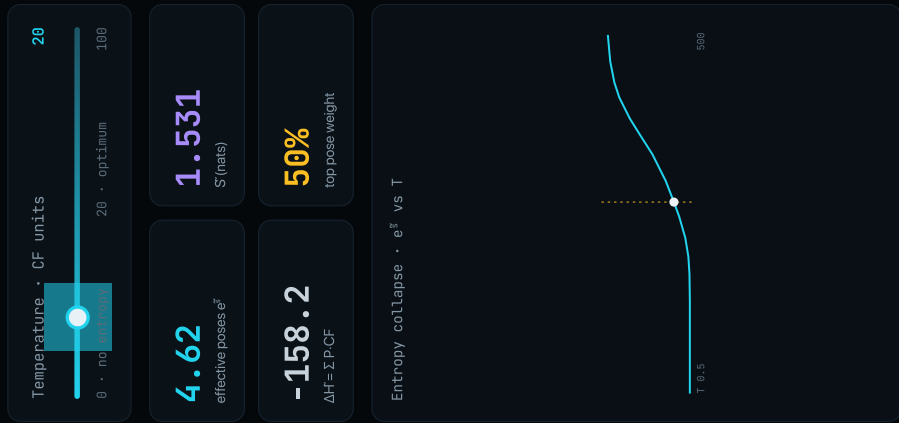
● **mode 2** wrong pose

3.71 Å from crystal

2 poses · best CF -142.49
RMSD to crystal 3.24-3.71 Å

— crystal — mode 1 — mode 2 ● N
● 0

// THE POPULATION, NOT THE POSE
20 poses. At T=20, 4.62 of them matter.



T optimised by Monte Carlo on PDBbind (mode 20, mean 21) · CF is dimensionless – T is not kelvin · T=0 collapses the ensemble to one pose

// ARCHITECTURE

Genetic algorithm in, ranked binding modes out.

01 · input

Atom typing

PDB / mmCIF receptor · MOL2 / SDF ligand · 40-
type NRGDock assignment

02 · search

Genetic algorithm

gaboom · rigid-body + torsion chromosomes · CF
selection pressure · parallel restarts

03 · score

Voronoi CF

contact areas → 40×40 energy matrix · soft-wall cap ·
all-pairs clash penalty

04 · elect

Mode election

cluster → G = H - T · ranking objective · two-gate spread
guard

StatMechEngine · diagnostic only

Canonical-form Z, F, Cv over CF samples — emitted as proxy_only.

tENCoM · diagnostic only

Torsional elastic-network vibrational scale — no kcal/mol calibration.

Runtime dispatch: CUDA → Metal → AVX-512 → AVX2 → OpenMP → scalar · selected at startup, no configuration

// SCORING · THE CONTACT FUNCTION

Shape complementarity from a Voronoi tessellation.

$CF = CF.com + CF.wal + CF.sas + CF.elec + CF.hbond + CF.pb_clash + CF.con$

40×40

energy matrix over SYBYL atom types, trained on PDB contact frequencies

820

unique interaction parameters after full symmetry

≤ 50

soft-wall cap in CF units per contact — no numerical blow-up

all-pairs

PoseBusters clash penalty on a loop-invariant receptor grid

Lower CF is better · a perfect complementary fit approaches the global minimum · full derivation in docs/SCORING.md

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// WHAT CHANGED VS FLEXAID

Five corrections between FlexAID and FlexAIDAS.

MECHANISM	FlexAID (2015)	FlexAIDAS
Ranking criterion	min(CF)	$\tilde{G} = \tilde{H} - T \cdot \tilde{S}$
False-minima demotion	x	two-gate spread guard
Intermolecular clash	23x undercount	full all-pairs penalty
Atom typing (N.2, N.3, C.1, I)	wrong sign / dead rows	corrected, live matrix rows
Soft-wall repulsion	unbounded (SIGSEGV)	capped at 50 CF units

Plus tENCov vibrational diagnostic (new) - every mechanism is an independently toggleable environment-variable flag for ablation

Same budget for every method, every target.

10

independent runs
per case, per method, per protocol

2,000,000

energy evaluations
per run — fixed budget, no method tuned per target

10,000

bootstrap iterations
median success rate, Efron 1981

< 2.0 Å

RMSD, top-10 poses
success criterion — top-10, not top-1

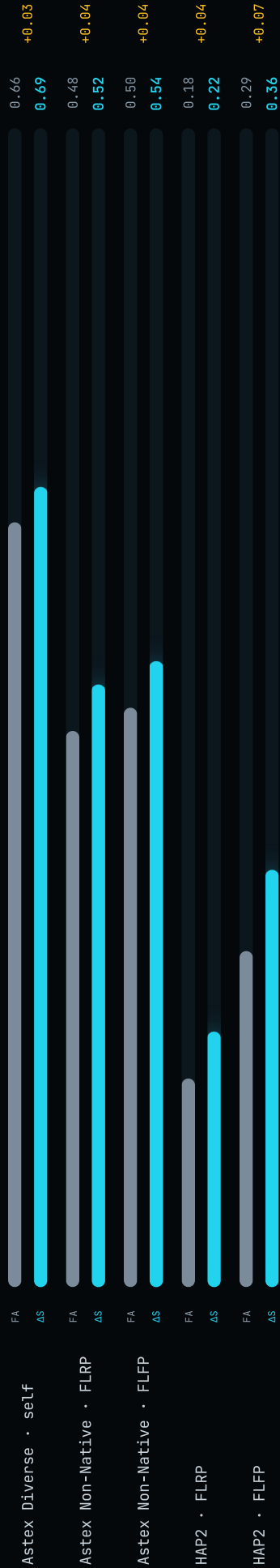
3

published, citable sets
a difficulty gradient, not a favourable subset

READ THIS RIGHT These are **top-10** success rates. Current campaigns report top-1 — the two are not comparable and never appear together. Open source, no licence gate, no private test set.

// COMPARISON · THE ΔS ABLATION

ΔS beats FlexAID in all five conditions tested.



■ FlexAID baseline (FA) ■ FlexAID ΔS (ΔS) gold = gain from ΔS

Same code, same budget, ΔS toggled · gains are largest where side-chain flexibility is decisive · ISMB/ECCB 2017 30sig

// ONE KERNEL, THREE DOMAINS

The same entropy math, ported three ways.

$H = -\sum p_i \ln p_i$ — compute it over any distribution, watch it collapse

FlexAIDAS

docking engine

Entropy over **GA pose ensembles** — the collapse tells a real binding funnel from a false minimum.

C++26 · Python · PyMOL plugin

Shannon

LLM safety referee

Entropy over **LLM token distributions** — detects entropy collapse in frontier agents at runtime.

C++28 · Python

NATuRaL

PokeDrug biofeedback

Entropy over **HRV intervals** — a Shannon Collapse Index scoring focus on Apple platforms.

SwiftUI · iOS / watchOS / visionOS

// WHAT'S NEXT

From proxy diagnostic to receipted benchmark.

01 Blind, receipted Astex-85

Unseeded campaign with binary digest, git object ID and runner command — the number becomes publishable only then.

02 ITC calibration to kcal/mol

Anchor CF-unit proxies to measured affinities so G can carry physical units, not just rank.

03 Methods manuscript

Morency & Najmanovich, 2026 — in preparation. Co-authors and independent reruns welcome.

04 Bindings & GPU dispatch

Harden the Python API; ship Metal / CUDA / AVX-512 backends selected automatically at startup.

// THE TAKE-HOME

Don't trust the numbers. Rerun them.

The score is not the asset — the protocol is. Three published sets, one fixed budget, a non-parametric interval, and an engine you can install. That is the part that transfers to your pipeline.

Fixed budget

2,000,000 × 10 runs

Identical for every method, every case. Nothing tuned per target.

Non-parametric

10,000 bootstrap

A median with a real interval, not a single lucky run.

Published sets

85 · 1112 · 76_N

Easy to hardest — a gradient we did not get to choose.

Open source / runs on a Laptop / no licence gate / no private test set

github.com/LeBonhommePharma

Najmanovich Research Group → Le Bonhomme Pharma.

Louis-Philippe Morency — Najmanovich Research Group, Université de Montréal; now open science at Le Bonhomme Pharma, Montréal.

PUBLICATIONS · L.-P. MORENCY

- 2018 Applications of the NRGsuite & FlexAID in computational drug discovery & design · *Methods Mol Biol* 1762, 367–388
 - 2018 Functional diversity of TMPRSS6 isoforms in hepatocellular carcinoma cell lines · *Sci Rep* 8, 12562
 - 2018 An alternative translation initiation site in the Glutathione S-Transferase tag · *J Biotechnol* 286, 14–16
 - 2017 Large-scale detection of drug off-targets for repurposing & side-effects · *BMC Pharmacol Toxicol* 18
 - 2015 NRGsuite — real-time FlexAID docking in PyMOL · *Bioinformatics* 31, 3856–3858
- Builds on FlexAID (Gaudreault & Najmanovich, JCTM 2015) and ENCoM (Frappier & Najmanovich, 2014), which tENCoM extends.

THE ASK · CO-AUTHOR

The methods manuscript is in preparation. Rerun the benchmarks
— then co-author them.

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